# ANISOMETROPIA

- Equql refraction in the two eyes is knows as “Isometropia”.

- When the total refraction of the two eyes is unequal the condition is know as “ansiometropia”.

- Small degree of anisometropia is of no concern.

- A difference of 1D in two eyes cause a 2% difference in the size of the two retinal images.

- A difference up to 5% in retinal images of two eyes is well tolerated.

- In other words, an anisometropia up to 2.5D is well tolerated and that between 2.5 and 4D can be tolerated depending upon the individual sensitivity.

- However, if it is more than 4D, it is often not tolerated and is a matter of concern.

## Etiology

1. **Congenital and developmental anisometropia :** It occurs due to differantial growth of the two eyeballs.

2. **Acquired anisometropia :** It may occur due to asymmetric age change, uniocular aphakia ( after cataract surgery ) or due to implantation of IOL of wrong power.

## Clinical Types

I. **Simple anisometropia :** In this one eye is normal (emmetropia) and the other eye is either myopic ( simple myopic anisometropia ) or hypermetropic ( simple hypermetropic anisometropia).

II. **Compound anisometropia :** where both eye are either hypermetropic ( compound hypermetropic anisometropic ) or myopic ( compound myopic anisometropic ).

-But one eye is having higher refractive error then the other.

III. **Mixed anisometropia :** In this one eye is myopic and the other hypermetropic. This is also called antimetropia.

IV. **Simple astigmatic anisometropia :** when one eye is normal and the other has either simple myopic or hypermetropic astigmatism.

V. **Compound astigmatic anisometropia :** when both eyes are astigmatic but of unequal degree.

## Status of binocular vision in anisometropia

i. **Binocular single vision :** It present in small degree of anisometropia ( < 3D ).

ii. **Uniocular vision :** when refractive error in one eye is of high degree ( ≥ 3D difference in two eyes ) that eye is suppressed and develops anisometropic amblyopia.
-Thus the patient has only uniocular vision.

iii. **Alternate vision :** It occurs when one eye is emmetropic or hypermetropic and the other myopic.
-The emmetropic or hypermetropic eye is used for distance vision and myopic eye is used for near vision.

**Diagnosis:** Diagnosis of anisometropia is made after retinoscopic examination or autorefractometry in patient with defective vision.

## Treatment:

1. **Spectacles :** The corrective spectacles can be tolerated up to a maximum difference of 2.5 D.
   -after that there may occur diplopia.

2. **Contact lenses :** are advised for higher degree of anisometropia.

3. **Aniseikonic glasses :** are also available but their clinical result are often disappointing.

4. **Refractive surgery for anisometropia include :**

- Intraocular lens ( IOL ) implantation for uniocular aphakia.
- Refractive cornral surgery ( LASIK ) for unilateral high myopia astigmatism and hypermetropia.
- Phakic refractive lenses ( PRL ) and refractive lens exchange ( RLE ) are quint useful in very high dergee ( < 7 D ) anisometropia.

**NOTE** - Efforts should be made to fully correct the ansiometropia in children to prevent anisometropia amblyopia.

In adults with amblyopia under correction of the more ametropic eye may be required to avoid ocular discomfart.